

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Assessment Tool Weightage: 25%

QUESTIONS:

Total Marks: 25

Case Study 1: CNN for Automated Medical Image Classification

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

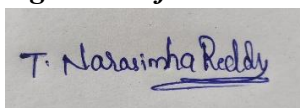
1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

Code of Conduct:

I T. Narasimha Reddy (192425190) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink that reads "T. Narasimha Reddy".

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

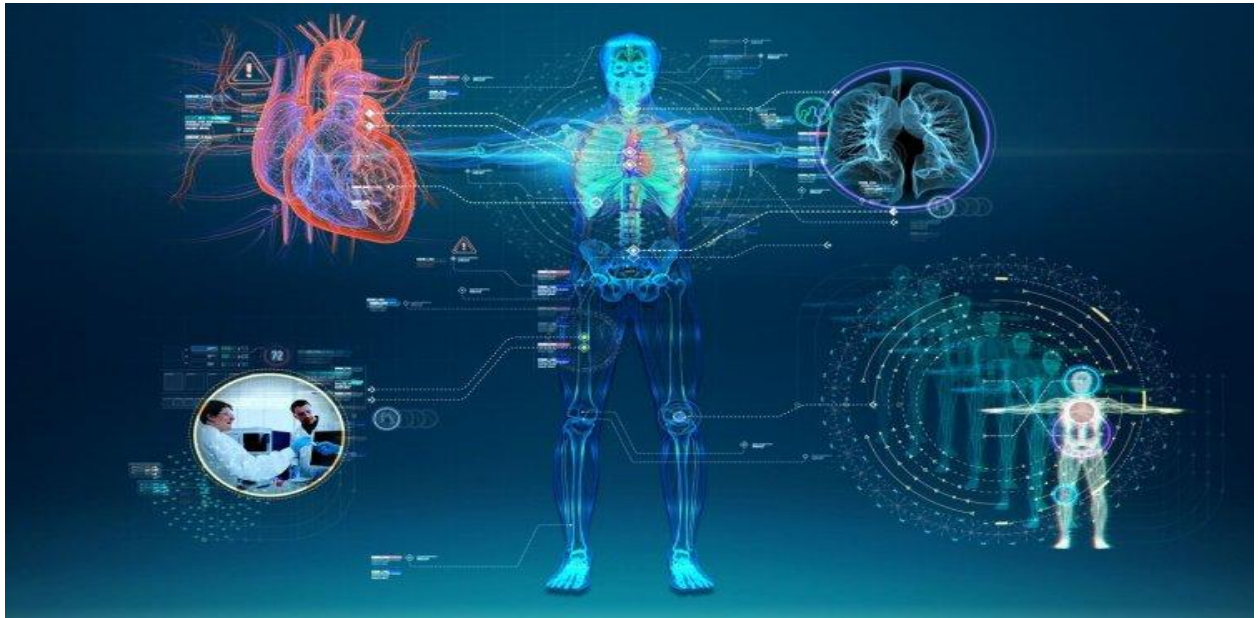
Case Study 1: CNN for Automated Medical Image Classification

1. Introduction

The healthcare organization aims to classify chest X-ray images into **Normal** and **Disease-Affected** categories. Since X-ray images may vary in size, brightness and quality, a **Convolutional Neural Network (CNN)** is suitable because it automatically learns important visual features such as edges, textures, shapes and abnormal regions.

2. Proposed CNN Architecture

A suitable architecture is:



Main Layers

- **Input Layer:** Accepts resized and normalized X-ray images.
- **Convolutional Layers:** Extract important image features.
- **ReLU Activation:** Introduces non-linearity.
- **Pooling Layers:** Reduce image dimensions and computation.
- **Batch Normalization:** Stabilizes and improves training.
- **Global Average Pooling:** Converts feature maps into a compact representation.
- **Dropout:** Reduces overfitting.
- **Fully Connected Layer:** Performs final classification.
- **Output Layer:** Uses sigmoid to classify the image as Normal or Disease-Affected.

Feature Learning

Image → Edges → Textures → Shapes → Abnormal Patterns → Classification

3. Motivation for Using CNN

CNNs are preferred because they are specifically designed for image processing.

Advantages

- Automatically extracts features from images.
- Preserves spatial relationships between pixels.
- Detects features at different locations.
- Requires less manual feature engineering.
- Learns hierarchical features from simple to complex patterns.
- Provides high classification performance for image-based tasks.

4. Role of Filters, Convolution, Pooling and Fully Connected Layers

Convolution and Filters

A **filter/kernel** is a small matrix, usually 3×3 , that slides across the image.

It detects features such as:

- Edges
- Lines
- Textures
- Shapes
- Abnormal patterns

The output of convolution is called a **feature map**.

Pooling

Pooling reduces the size of feature maps while retaining important information.

Max pooling selects the maximum value from a small region.

Benefits:

- Reduces computation
- Reduces memory usage
- Helps control overfitting
- Provides some translation tolerance

Fully Connected Layer

The fully connected layer uses the extracted features to make the final classification.

Output:

Normal $\rightarrow 0$

Disease-Affected $\rightarrow 1$

5. Parameter Sharing

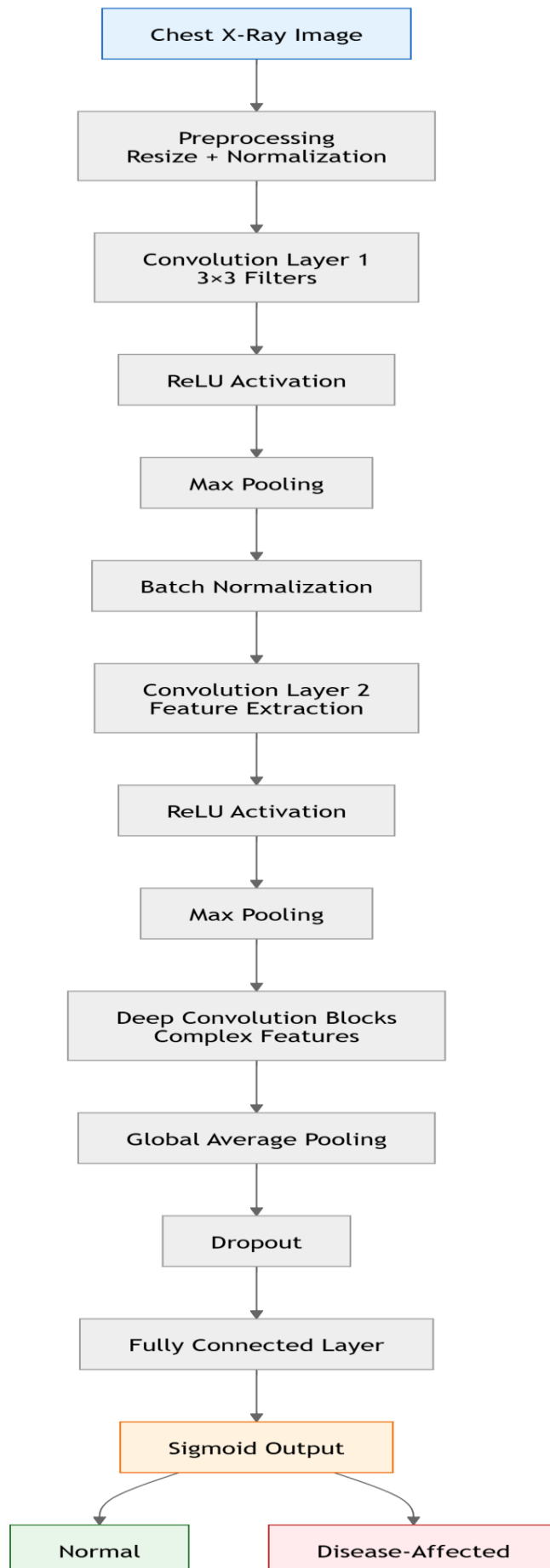
CNNs use the **same filter weights at different locations** of an image. This is called parameter sharing.

For example, a 3×3 filter requires only **9 weights + 1 bias**, and the same filter can scan the complete image.

In contrast, a fully connected network requires separate weights for individual pixel connections.

Benefits

- Greatly reduces the number of parameters.
- Reduces memory and computational requirements.
- Helps detect the same feature at different locations.
- Improves generalization.



6. Regularization Techniques

Medical image datasets can cause overfitting if the model memorizes training images instead of learning general patterns.

Data Augmentation

Images can be modified using suitable:

- Rotation
- Scaling
- Cropping
- Brightness/contrast changes

This increases image diversity.

Dropout

Dropout randomly disables some neurons during training, preventing the network from depending too much on specific neurons.

Batch Normalization

Batch normalization normalizes intermediate activations and makes training more stable and faster.

Other Techniques

- Early stopping
- L2 regularization
- Learning-rate scheduling

7. AlexNet vs ResNet

Feature	AlexNet	ResNet
Depth	Relatively shallow	Can be very deep
Feature learning	Good	Strong
Skip connections	No	Yes
Vanishing gradient	More susceptible	Reduced
Training deep networks	Difficult	Easier
Performance	Good	Generally better
Medical application	Suitable	More suitable

Recommended Architecture: ResNet

ResNet is preferred because its residual/skip connections make deep networks easier to train and help overcome the vanishing-gradient problem.

It can learn complex features from X-ray images and generally provides stronger feature representations than AlexNet.

For practical implementation, a moderate model such as **ResNet-18 or ResNet-34** can provide a good balance between accuracy and computational requirements.

flowchart TD

Conclusion

CNNs are highly suitable for automated chest X-ray classification because they automatically learn meaningful visual features. **Convolution and filters** perform feature extraction, **pooling** reduces computational complexity, **parameter sharing** reduces the number of trainable parameters, and **dropout, augmentation and batch normalization** help prevent overfitting. Among AlexNet and ResNet, **ResNet is the better choice** because of its deeper feature-learning capability and residual connections.

Case Study 2: CNN-Based Autonomous Vehicle Object Recognition

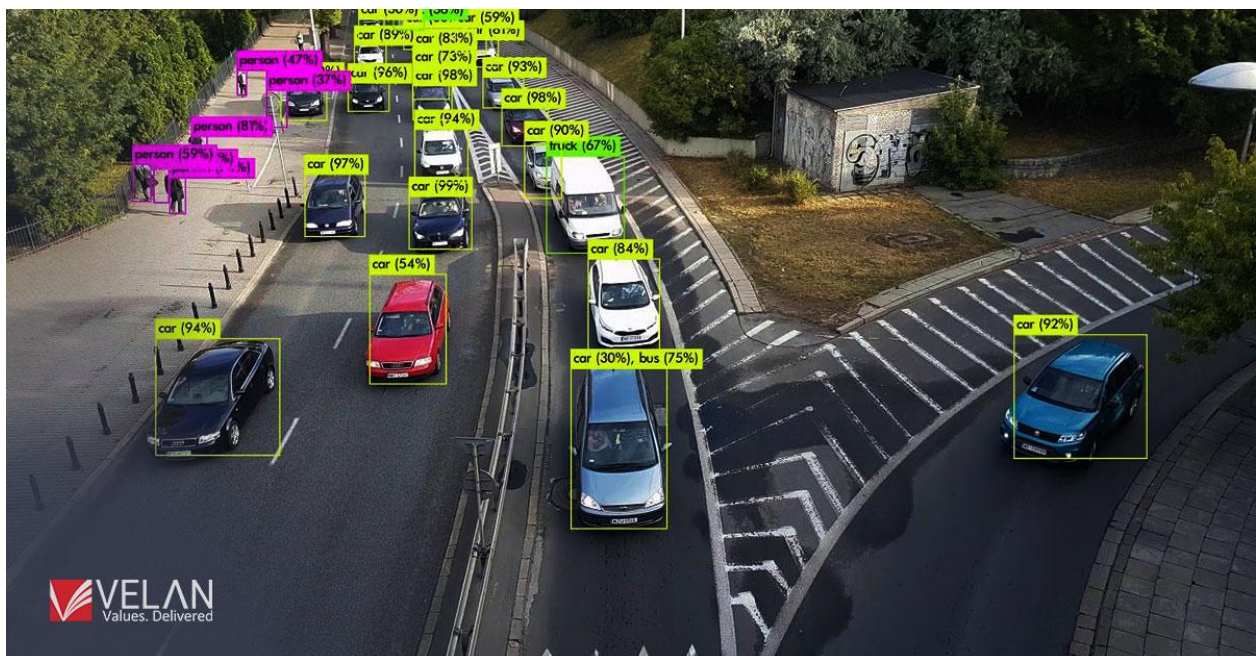
1. Introduction

An autonomous vehicle must identify objects such as **cars, pedestrians, bicycles, traffic signs and road obstacles** from camera images. Since the environment changes continuously due to lighting, distance, weather and viewing angle, a robust CNN-based vision system is required.

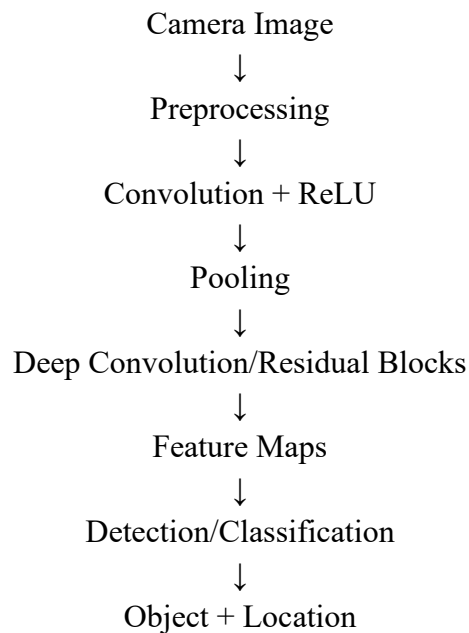
2. Proposed CNN Architecture

A suitable architecture is:

The system can be designed using a **ResNet backbone** for feature extraction.



Basic Flow



For autonomous vehicles, **object detection** is more appropriate than simple classification because the system needs to identify **what the object is and where it is located**.

3. Role of CNN Layers

Input Layer

Receives camera images and performs preprocessing such as:

- Resizing
- Normalization
- Noise reduction

Convolutional Layer

Uses filters to detect visual patterns.

Early layers detect:

- Edges
- Lines
- Corners

Deeper layers detect:

- Wheels
- Vehicles
- Human shapes
- Traffic signs
- Road obstacles

ReLU Activation

Adds non-linearity and allows the network to learn complex patterns.

Pooling

Reduces spatial dimensions and computational requirements while retaining important features.

Output Layer

Produces object classes and, in an object-detection system, their locations using bounding boxes.

4. Progressive Feature Learning

CNN filters learn features progressively.

Pixels

↓

Edges and Lines

↓

Textures and Shapes

↓

Object Parts

↓

Complete Objects

↓

Object Recognition

For example:

Early layer → Edge of a car

Middle layer → Wheel/window

Deep layer → Complete car

This hierarchical learning allows CNNs to recognize complex objects.

5. Parameter Sharing

The same convolutional filter is applied across different locations of the image.

For example, a filter that detects a vertical edge can detect that edge whether it occurs on the **left, center or right side** of the image.

Benefits

- Reduces trainable parameters.
- Reduces memory and computational cost.
- Allows features to be detected at different locations.
- Improves translation-related feature detection.
- Makes CNNs suitable for large camera images.

6. Overfitting and Regularization

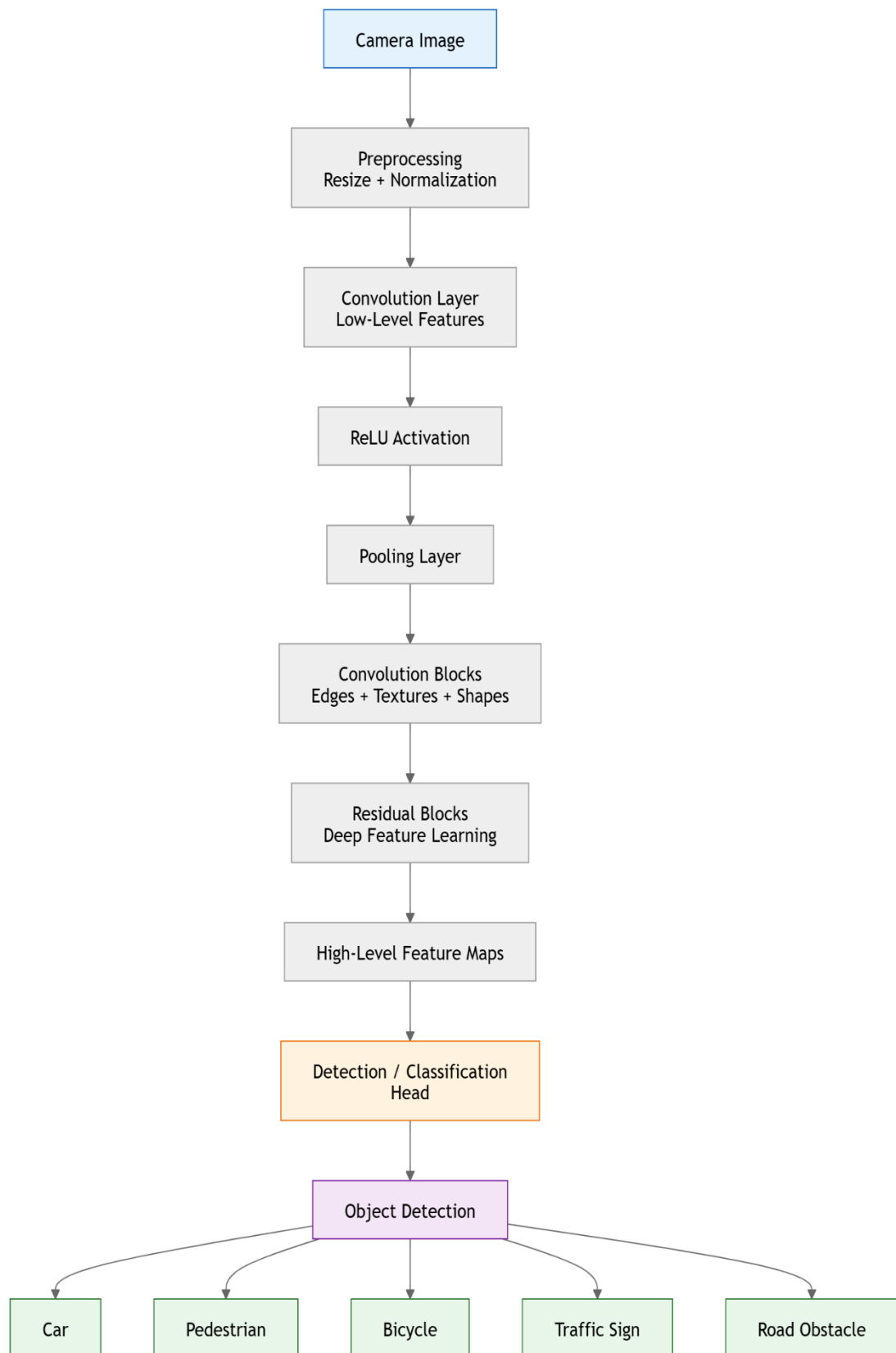
Overfitting can occur when the model performs well on training images but fails on new road conditions.

Suitable techniques

Data Augmentation

Use realistic variations such as:

- Brightness changes
- Scaling
- Cropping
- Rotation



Dropout

Randomly removes neurons during training and reduces dependency on specific features.

Batch Normalization

Improves training stability and convergence.

Early Stopping

Stops training when validation performance stops improving.

Diverse Dataset

Training data should include:

- Day and night
- Rain and clear weather
- Urban and rural roads
- Different camera angles
- Different object distances

7. AlexNet vs ResNet

Feature	AlexNet	ResNet
Architecture	Conventional CNN	Residual CNN
Depth	Shallow	Deep
Skip connections	No	Yes
Vanishing gradient	More likely	Reduced
Feature learning	Good	Excellent
Training deep networks	More difficult	Easier
Accuracy	Good	Generally higher
Real-time use	Lower complexity	Depends on model size

8. Recommended Architecture

ResNet is recommended for the autonomous vehicle system.

Reasons

High Accuracy:

ResNet can learn complex features required to distinguish pedestrians, vehicles, signs and obstacles.

Better Feature Learning:

Deep residual networks learn high-level semantic features effectively.

Reduced Vanishing Gradient:

Skip connections allow information and gradients to flow through deeper layers.

Real-Time Consideration:

A moderate model such as **ResNet-18** or **ResNet-34** can provide a better balance between accuracy and computational cost than an unnecessarily deep model.

Detection Capability:

ResNet can be used as a feature-extraction backbone for an object-detection system.

Conclusion

CNNs provide an effective solution for autonomous vehicle vision because they automatically learn hierarchical features from camera images. **Convolution and filters** identify visual patterns, **pooling** reduces computation, and **parameter sharing** enables efficient feature detection across different image locations.

Although AlexNet is computationally simpler, **ResNet is generally more suitable** because it provides stronger feature learning and reduces vanishing-gradient problems. A moderate ResNet model can achieve a practical balance between **accuracy, computational cost and real-time performance**.